

Switching, Routing, and Wireless Essentials

Introduction

SM Networks LLC is a network solutions provider headquartered in Vancouver, Canada having branch offices in Burnaby and Surrey. All three locations access the internet through their respective ISP. The connection to the internet is represented by Google Server connected to the Internet.

Objectives

The objective of this lab is to configure the partially configured network and establish full network connectivity among corporate branch offices and internet as per prescribed policy.

Instructions

You have been asked to prototype the network in Packet Tracer for evaluation by senior network staff. For the sake of time, many repetitive, but important, configuration tasks have been omitted from this activity. Many of these tasks, especially those related to security, are essential elements of network configuration. The intent of this activity is not to diminish the importance of full device configurations. All initial router password for User EXEC mode is set to **cisco** and Privileged EXEC mode is set to **ccna**.

Part 1: IP Addressing Scheme

In this activity, the following network address space has been utilized to provide IP addressing for the network depicted in the topology diagram. VLSM is being used so that the addressing requirements can be met while wasting the least amount of address space. The Vancouver network operates with dual-stack support, utilizing both IPv4 and IPv6 protocols. In contrast, the branch networks are currently limited to IPv4, with IPv6 deployment still in the planning phase.

VLAN and Host Information			
Vancouver Network			
VLAN ID	VLAN Name	Network Address	Host IP Addresses
1	default	172.16.196.0/27	Assign the 1 st IP to TFTP Server
		2001:211:BC:1::/64	Assign the 10 th IP to TFTP Server
2	Sales	172.16.192.0/23	Host addresses should be dynamically configured
		2001:211:BC:2::/64	Host addresses should be dynamically configured
3	Marketing	172.16.194.128/25	Assign the 3 rd IP to the Marketing PC
		2001:211:BC:3::/64	Assign the 11 th IP to the Marketing PC
4	Administration	172.16.196.32/27	Assign the 4 th IP to the Admin PC
		2001:211:BC:4::/64	Assign the 14 th IP to the Admin PC
5	IT	172.16.196.96/29	Assign the 5 th IP to the IT PC
		2001:211:BC:5::/64	Assign the 5 th IP to the IT PC
Burnaby Network			
1	default	172.16.201.0/24	Assign the 100 th and 200 th IP address to the Branch Admin and Radius Server respectively. Assign the last usable IP address to the Burnaby router gateway interface.
2	WLAN-Manage	172.16.202.0/24	Assign the last IP to the WLC Management interface
3	WLAN-Employee	172.16.203.0/24	Host addresses should be dynamically configured
4	WLAN-Guest	172.16.204.0/24	Host addresses should be dynamically configured
Surrey Network			
1	default	172.16.208.0/24	Assign the last 2 usable IP addresses to the GW1 and GW2 respectively. Assign 1 st IP to Surrey PC.

Link Information				
ISP to Vancouver (pre-configured)	ISP	Serial 0/1/0	142.31.216.1	2001:211:DBA:1::1
	Vancouver	Serial 0/0/0	142.31.216.2	2001:211:DBA:1::2
ISP to Burnaby (pre-configured)	ISP	Serial 0/1/1	24.207.53.133	N/A
	Burnaby	Serial 0/1/0	24.207.53.133	N/A
ISP to Surrey GW1 (pre-configured)	ISP	Serial 0/2/0	154.5.19.1	N/A
	Surrey GW1	Serial 0/1/0	154.5.19.2	N/A
ISP to Surrey GW2 (pre-configured)	ISP	Serial 0/2/1	154.5.19.5	N/A
	Surrey GW2	Serial 0/1/0	154.5.19.6	N/A

Part 2: Configure VLAN, Etherchannel and Inter-VLAN Routing

1. Configure VLAN in the Vancouver network. Use the VLAN table to create and name the VLANs on the appropriate switches. VLAN names are case-sensitive.

```
DLSW-A(config)#vlan 2
DLSW-A(config-vlan)#name Sales
DLSW-A(config-vlan)#vlan 3
DLSW-A(config-vlan)#name Marketing
DLSW-A(config-vlan)#vlan 4
DLSW-A(config-vlan)#name Administration
DLSW-A(config-vlan)#vlan 5
DLSW-A(config-vlan)#name IT
DLSW-A(config-vlan)#end
DLSW-A#show vlan brief
```

Similarly create VLANs in ALSW as we are not using VTP here.

```
ALSW-A(config)#vlan 2
ALSW-A(config-vlan)#name Sales
ALSW-A(config-vlan)#vlan 3
ALSW-A(config-vlan)#name Marketing
ALSW-A(config-vlan)#vlan 4
ALSW-A(config-vlan)#name Administration
ALSW-A(config-vlan)#vlan 5
ALSW-A(config-vlan)#name IT
ALSW-A(config-vlan)#exit
ALSW-A#show vlan brief
```

2. Configure the appropriate ports that link the switches as functioning trunk ports. Use the `range` command to select the appropriate ports.

```
DLSW-A(config)#int range Gig1/0/1-2, Gig1/0/24
DLSW-A(config-if-range)#switchport mode trunk
DLSW-A(config-if-range)#end
DLSW-A#show interfaces trunk
```

```
ALSW-A(config)#int range Gig0/1-2
ALSW-A(config-if-range)#switchport mode trunk
ALSW-A(config-if-range)#exit
ALSW-A#show interfaces trunk
```

3. Identify the required VLAN for each switch port based on the connected devices. Assign each port to its respective VLAN and ensure it is configured as an access port.

```
ALSW-A(config)#int f0/1
ALSW-A(config-if)#switchport mode access
ALSW-A(config-if)#switchport access vlan 2
```

```

ALSW-A(config-if)#exit
ALSW-A(config)#int f0/2
ALSW-A(config-if)#switchport mode access
ALSW-A(config-if)#switchport access vlan 3
ALSW-A(config-if)#exit
ALSW-A(config)#int f0/3
ALSW-A(config-if)#switchport mode access
ALSW-A(config-if)#switchport access vlan 1
ALSW-A(config-if)#exit
ALSW-A(config)#int f0/4
ALSW-A(config-if)#switchport mode access
ALSW-A(config-if)#switchport access vlan 4
ALSW-A(config-if)#exit
ALSW-A(config)#int f0/5
ALSW-A(config-if)#switchport mode access
ALSW-A(config-if)#switchport access vlan 5
ALSW-A(config-if)#exit

```

4. Create EtherChannel according to the EtherChannel Port Assignments Table. Both sides of the channel should attempt to negotiate the link protocol. Configure the component physical interfaces and port channel interfaces as static trunks.

Etherchannel Port Assignments Table			
Channel Group	Protocol to be Used	Devices in Groups	Ports in Group
1	LACP	DLSW-A	Gig1/0/1, Gig1/0/2
		ALSW-A	Gig0/1, Gig0/2

```

DLSW-A(config)#int range Gig1/0/1-2
DLSW-A(config-if-range)#channel-group 1 mode active
DLSW-A(config-if-range)#exit
DLSW-A(config)#interface Port-channel 1
DLSW-A(config-if)#switchport mode trunk
DLSW-A(config-if)#exit

```

```

ALSW-A(config)#int range Gig0/1-2
ALSW-A(config-if-range)#channel-group 1 mode active
ALSW-A(config-if-range)#exit
ALSW-A(config)#int Port-channel 1
ALSW-A(config-if)#switchport mode trunk
ALSW-A(config-if)#exit

```

```

DLSW-A#show etherchannel summary
DLSW-A#show interface status

```

5. Configure Dual-Stack (IPv4 and IPv6) router-on-a-stick Inter-VLAN Routing in Vancouver router to allow seamless inter-VLAN communication across both protocols.
 - a. Use the **VLAN numbers** as the interface numbers.
 - b. For IPv4, assign the **last usable address** in the VLAN network to the respective interface.
 - c. For IPv6, assign the address **2001:211:BC:<VLAN>::1** to the corresponding VLAN interface within the VLAN network.
 - d. Manually configure a unique link-local address on each sub-interface in the format:
FE80::<sub-interface number>

```

Vancouver(config)#int g0/1.1
Vancouver(config-subif)#encapsulation dot1Q 1
Vancouver(config-subif)#ip address 172.16.196.30 255.255.255.224
Vancouver(config-subif)#ipv6 address fe80::1 link-local
Vancouver(config-subif)#ipv6 address 2001:211:BC:1::1/64

```

```

Vancouver(config-subif)#exit
Vancouver(config)#int g0/1.2
Vancouver(config-subif)#encapsulation dot1Q 2
Vancouver(config-subif)#ip address 172.16.193.254 255.255.254.0
Vancouver(config-subif)#ipv6 address fe80::2 link-local
Vancouver(config-subif)#ipv6 address 2001:211:BC:2::1/64
Vancouver(config-subif)#exit
Vancouver(config)#int g0/1.3
Vancouver(config-subif)#encapsulation dot1Q 3
Vancouver(config-subif)#ip address 172.16.194.254 255.255.255.128
Vancouver(config-subif)#ipv6 address fe80::3 link-local
Vancouver(config-subif)#ipv6 address 2001:211:BC:3::1/64
Vancouver(config-subif)#exit
Vancouver(config)#int g0/1.4
Vancouver(config-subif)#encapsulation dot1Q 4
Vancouver(config-subif)#ip address 172.16.196.62 255.255.255.224
Vancouver(config-subif)#ipv6 address fe80::4 link-local
Vancouver(config-subif)#ipv6 address 2001:211:BC:4::1/64
Vancouver(config-subif)#exit
Vancouver(config)#int g0/1.5
Vancouver(config-subif)#encapsulation dot1Q 5
Vancouver(config-subif)#ip add 172.16.196.102 255.255.255.248
Vancouver(config-subif)#ipv6 address fe80::5 link-local
Vancouver(config-subif)#ipv6 address 2001:211:BC:5::1/64
Vancouver(config-subif)#exit
Vancouver(config)#int g0/1
Vancouver(config-if)#no shutdown
Vancouver(config-if)#exit

```

```

Vancouver(config)#show ip interface brief
Vancouver(config)#show ipv6 interface brief
Vancouver(config)#show ip route
Vancouver(config)#show ipv6 route

```

6. Assign IPv4 and IPv6 addresses to all hosts in the Vancouver network as specified in the Host and VLAN Information table. Set Google's Public DNS IP addresses (8.8.8.8 for IPv4 and 2001:4860:4860::8888 for IPv6) as the DNS servers. Click Fast Forward Time to speed up the process. Verify connectivity between the hosts and save a backup copy of the router configuration to the TFTP server. Use default file name.
7. VLANs are pre-created on the Burnaby network switches. Configure functioning trunk ports and access ports as necessary. Add interface descriptions for easy port identification. Set the **VLAN-Manage** VLAN as the Native VLAN.

```

DLSW-B(config)#int Gig1/0/1
DLSW-B(config-if)#switchport mode trunk
DLSW-B(config-if)#description Connection to Access Switch
DLSW-B(config-if)#switchport trunk native vlan 2
DLSW-B(config-if)#exit

```

```

ALSW-B(config)#int g0/1
ALSW-B(config-if)#description Connection to Multi-Layer Switch
ALSW-B(config-if)#switchport mode trunk
ALSW-B(config-if)#switchport trunk native vlan 2
ALSW-B(config-if)#exit
ALSW-B(config)#int g0/2
ALSW-B(config-if)#description Connection to WLC-1
ALSW-B(config-if)#switchport mode trunk
ALSW-B(config-if)#switchport trunk native vlan 2
ALSW-B(config-if)#exit

```

```

ALSW-B(config)#int fa0/3
ALSW-B(config-if)#description Connection to LAP-1
ALSW-B(config-if)#switchport mode trunk
ALSW-B(config-if)#switchport trunk native vlan 2
ALSW-B(config-if)#exit
ALSW-B(config)#int range f0/1-2
ALSW-B(config-if-range)#switchport mode access
ALSW-B(config-if-range)#switchport access vlan 1
ALSW-B(config-if-range)#exit

```

8. Configure inter-VLAN routing on Burnaby Multi-Layer Switch to allow seamless inter-VLAN communication between different VLANs. Use the VLAN numbers for the required interface numbers. Use the first addresses in the VLAN networks for the interfaces.

```

DLSW-B(config)#ip routing

DLSW-B(config)#int vlan 1
DLSW-B(config-if)#ip address 172.16.201.1 255.255.255.0
DLSW-B(config-if)#no shutdown
DLSW-B(config-if)#exit
DLSW-B(config)#int vlan 2
DLSW-B(config-if)#ip address 172.16.202.1 255.255.255.0
DLSW-B(config-if)#exit
DLSW-B(config)#int vlan 3
DLSW-B(config-if)#ip address 172.16.203.1 255.255.255.0
DLSW-B(config-if)#exit
DLSW-B(config)#int vlan 4
DLSW-B(config-if)#ip address 172.16.204.1 255.255.255.0
DLSW-B(config-if)#exit

```

9. Assign host IP addresses in the Burnaby network according to the VLAN and Host Information table.

Part 3: Configure DHCP

1. Configure DHCPv4 in the Vancouver router for dynamically assigning IPv4 address to the sales department. Create a DHCP pool named **Sales**. Exclude the last 5 usable IP addresses of the subnet from the pool. Configure Google's public DNS IPv4 address 8.8.8.8 as the DNS server.

```

Vancouver(config)#ip dhcp excluded-address 172.16.193.250 172.16.193.254
Vancouver(config)#ip dhcp pool Sales
Vancouver(dhcp-config)#network 172.16.192.0 255.255.254.0
Vancouver(dhcp-config)#default-router 172.16.193.254
Vancouver(dhcp-config)#dns-server 8.8.8.8
Vancouver(dhcp-config)#exit

```

```

Vancouver#show ip dhcp pool
Vancouver#show ip dhcp binding

```

2. Configure Stateful DHCPv6 in the Vancouver router for dynamically assigning IPv6 address to the Sales department. Create a DHCP pool named **Sales-DHCPv6**. Configure Google's public DNS IPv6 address 2001:4860:4860::8888 as the DNS server.

```

Vancouver(config)#ipv6 dhcp pool Sales-DHCPv6
Vancouver(config-dhcpv6)#address prefix 2001:211:BC:2::/64
Vancouver(config-dhcpv6)#dns-server 2001:4860:4860::8888
Vancouver(config-dhcpv6)#exit
Vancouver(config)#int g0/1.2
Vancouver(config-subif)#ipv6 dhcp server Sales-DHCPv6

```

```
Vancouver(config-subif)#ipv6 nd managed-config-flag
Vancouver(config-subif)#end
```

3. Configure Stateless DHCPv6 in the Vancouver router for dynamically assigning IPv6 address to the Marketing department. Create a DHCP pool named **Marketing-DHCPv6**. Configure Vancouver network Server address as the DNS server address.

```
Vancouver(config)#ipv6 dhcp pool Marketing-DHCPv6
Vancouver(config-dhcpv6)#dns-server 2001:211:BC:1::9
Vancouver(config-dhcpv6)#exit
Vancouver(config)#int g0/1.3
Vancouver(config-subif)#ipv6 dhcp server Marketing-DHCPv6
Vancouver(config-subif)#ipv6 nd other-config-flag
Vancouver(config-subif)#end
```

```
Vancouver#show ipv6 dhcp pool
Vancouver#show ipv6 dhcp binding
```

Part 4: Configure Switch Security

1. Enforce Switchport security feature in ALSW-A.
 - a. Configure the active ports to accept a maximum of 2 MAC addresses.
 - b. The learned MAC addresses should be recorded in the running configuration.
 - c. If a security violation occurs, the switch ports should provide notification that a violation has occurred and place the interface in an err-disabled state.
 - d. MAC addresses should be present in the MAC address table for a maximum of 10 minutes before they are removed.

```
ALSW-A(config)#int range f0/1-5
ALSW-A(config-if-range)#switchport port-security
ALSW-A(config-if-range)#switchport port-security maximum 2
ALSW-A(config-if-range)#switchport port-security mac-address sticky
ALSW-A(config-if-range)#switchport port-security violation shutdown
ALSW-A(config-if-range)#switchport port-security aging time 10
ALSW-A(config-if-range)#exit
```

2. Guard against ARP attacks by implementing DAI
 - a. Activate DAI globally in ALSW-A. Configure DAI for the configured VLANs.

```
ALSW-A(config)#ip arp inspection vlan 1-5
ALSW-A(config-if-range)#ip arp inspection trust
ALSW-A(config-if-range)#exit
```

3. Mitigate STP attacks.

- a. Configure Rapid-PVST on all switches in Vancouver network.

```
DLSW-A(config)#spanning-tree mode rapid-pvst
ALSW-A(config)#spanning-tree mode rapid-pvst
```
- b. Configure BPDUguard and PortFast on the active ports of ALSW-A.

```
ALSW-A(config)#int range f0/1-5
ALSW-A(config-if-range)#spanning-tree portfast
ALSW-A(config-if-range)#spanning-tree bpduguard enable
ALSW-A(config-if-range)#exit
```
- c. Secure unused switch ports
 - a. On ALSW-A, move all unused switch ports to VLAN 999.
 - b. Configure all unused switch ports as static access ports.
 - c. Deactivate all unused switch ports.

```
ALSW-A(config)#vlan 999
```

```

ALSW-A(config-vlan)#exit
ALSW-A(config)#int range f0/6-24
ALSW-A(config-if-range)#switchport mode access
ALSW-A(config-if-range)#switchport access vlan 999
ALSW-A(config-if-range)#shutdown
ALSW-A(config-if-range)#exit

```

Part 5: Configure Static Routing

1. Configure IPv4 and IPv6 default routes where necessary to direct traffic for unknown destinations. Use Exit Interface arguments for IPv4 routes and Use Next-Hop Address arguments for IPv6 routes.

```

Vancouver(config)#ip route 0.0.0.0 0.0.0.0 Serial0/0/0
Vancouver(config)#ipv6 route ::/0 2001:211:DBA:1::1

```

```

Burnaby(config)#ip route 0.0.0.0 0.0.0.0 Serial0/1/0

```

```

Surrey-GW1(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
Surrey-GW1(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0

```

Part 6: Configure FHRP

1. Configure HSRP for redundancy in the Surrey network.
 - a. Configure Gateway IP address on the physical interfaces of Surrey-GW1 and Surrey-GW2.


```

Surrey-GW1(config)#int g0/0
Surrey-GW1(config-if)#ip address 172.16.208.254 255.255.255.0
Surrey-GW1(config-if)#no shutdown

Surrey-GW2(config)#int g0/0
Surrey-GW2(config-if)#ip address 172.16.208.253 255.255.255.0
Surrey-GW2(config-if)#no shutdown

```
 - b. Configure GW1 as the Active and GW2 as the Standby Router. Set 172.16.208.252 as the Standby IP. For Active Router, keep the default priority and for Standby Router set the priority to 150. Configure HSRP Preemption.

```

Surrey-GW1(config)#int g0/0
Surrey-GW1(config-if)#standby 1 ip 172.16.208.252
Surrey-GW1(config-if)#standby 1 preempt
Surrey-GW1(config-if)#exit

```

```

Surrey-GW2(config)#int g0/0
Surrey-GW2(config-if)#standby 1 ip 172.16.208.252
Surrey-GW2(config-if)#standby 1 priority 150
Surrey-GW2(config-if)#standby 1 preempt
Surrey-GW2(config-if)#exit

```

```

Surrey-GW1#show standby

```

Part 7: Configure a Wireless LAN using a Wireless LAN Controller

In this part of the assessment, you will configure the wireless LAN controller to provide wireless access to the Burnaby network. Username and password are the default **admin/Admin123**.

1. On the Burnaby multi-layer switch, configure IPv4 DHCP scopes for the WLAN networks: **WLAN-Employee** and **WLAN-Guest**. Exclude the first two IP addresses from each pool. Ensure the pools provide the default gateway and DNS server addresses.

```

DLSW-B(config)#ip dhcp excluded-address 172.16.203.1 172.16.203.2
DLSW-B(config)#ip dhcp pool WLAN-Employee

```

```

DLSW-B(dhcp-config)#network 172.16.203.0 255.255.255.0
DLSW-B(dhcp-config)#default-router 172.16.203.1
DLSW-B(dhcp-config)#dns-server 8.8.8.8
DLSW-B(dhcp-config)#exit
DLSW-B(config)#ip dhcp excluded-address 172.16.204.1 172.16.204.2
DLSW-B(config)#ip dhcp pool WLAN-Guest
DLSW-B(dhcp-config)#network 172.16.204.0 255.255.255.0
DLSW-B(dhcp-config)#default-router 172.16.204.1
DLSW-B(dhcp-config)#dns-server 8.8.8.8
DLSW-B(dhcp-config)#exit

```

- On the Burnaby multi-layer switch, configure a DHCP scope for the Management Network named **WLAN-Manage**. This will be used by the LAPs and other management devices on the network. Exclude the router interface address and WLC management interface address.

```

DLSW-B(config)#ip dhcp excluded-address 172.16.202.1
DLSW-B(config)#ip dhcp excluded-address 172.16.202.254
DLSW-B(config)#ip dhcp pool WLAN-Manage
DLSW-B(dhcp-config)#network 172.16.202.0 255.255.255.0
DLSW-B(dhcp-config)#default-router 172.16.202.1
DLSW-B(dhcp-config)#dns-server 8.8.8.8
DLSW-B(dhcp-config)#exit

```

- Connect to the WLC over HTTPS to the management interface. Username and password are the default **admin/Admin123**.
- Configure the WLC with the RADIUS server IPv4 address. Use a shared secret of **RADsecret**.
Security → New → Provide Server IP Address and Shared Secret → Apply → Save Configuration
Configure Radius Server to authenticate WLAN users. The RADIUS server will authenticate the WLC before it allows the WLC to access the user account information that is on the server. This requires a shared secret value. The Radius server is pre-configured with few usernames and passwords for test purposes.

AAA

Service ☒ On ☐ Off		Radius Port 1812	
---	--	---	--

Network Configuration

Client Name	Client IP
Secret	ServerType Radius

	Client Name	Client IP	Server Type	Key
1	WLC-1	172.16.202.254	Radius	RADsecret

User Setup

Username	Password
-------------------------------	-------------------------------

	Username	Password
1	umme	salsabil

- Create a new interface named **WLAN-3**. The interface should use **VLAN 3** and physical port 1. Use the last IP address of the subnet as Interface Address. Multiple VLAN interfaces can use the same physical port because the physical interfaces are like dedicated trunk ports.

Controller → Interfaces → New → Interface Name: WLAN-3, VLAN Id: 3 → Apply

Physical Information

Port Number: 1

Interface Address

VLAN Identifier: 3

IP Address: 172.16.203.2

Netmask: 255.255.255.0

Gateway: 172.16.203.1

DHCP Information

Primary DHCP Server: 172.16.203.1

Apply → Save Configuration

6. Create another new interface named **WLAN-4**. The interface should use **VLAN 4** and physical port 1. Use the last IP address of the subnet as Interface Address.

Controller → Interfaces → New → Interface Name: WLAN-4, VLAN Id: 4 → Apply

Physical Information

Port Number: 1

Interface Address

VLAN Identifier: 4

IP Address: 172.16.204.2

Netmask: 255.255.255.0

Gateway: 172.16.204.1

DHCP Information

Primary DHCP Server: 172.16.204.1

Apply → Save Configuration

7. Create a new WLAN for branch office users.
 - a. Use the profile name **SMNET-Employee**. Assign an SSID of **SMNET**. The wireless LAN should use the VLAN interface **WLAN-3** that was previously configured.
WLANs → Create New → Profile Name: SMNET-Employee, SSID: SMNET → Apply
 - b. Configure the WLAN to use the WPA2 security policy and dot1x Authentication Key Management.
General → Status: Enabled, Interface/Interface Group(G): WLAN-3
Security → Layer 2 Security: WPA+WPA2, WPA+WPA2 Parameters: WPA2 Policy, 802.1X: Enable
 - c. Configure the WLAN to use the RADIUS server that was previously configured to authenticate wireless users.
Security → AAA Servers → Server 1: IP 172.16.201.200 Port 1812
 - d. Activate **FlexConnect Local Switching** and **FlexConnect Local Auth**.
Advanced → FlexConnect → FlexConnect Local Switching: Enabled, FlexConnect Local Auth: Enabled

Apply → Save Configuration

8. Create another WLAN for guest users.
 - a. Use the profile name **SMNET-Guest**. Assign an SSID of **SMNET-GUEST**. The wireless LAN should use the VLAN interface **WLAN-4** that was previously configured.
WLANs → Create New → Profile Name: SMNET-Guest, SSID: SMNET-GUEST → Apply
 - b. Configure the WLAN to use the WPA2 security policy and PSK Authentication Key Management. Use passphrase **SMNET2024**.
General → Status: Enabled, Interface/Interface Group(G): WLAN-4
Security → Layer 2 Security: WPA+WPA2, WPA+WPA2 Parameters: WPA2 Policy, PSK: Enabled
PSK Format: SMNET2024
 - c. Activate **FlexConnect Local Switching** and **FlexConnect Local Auth**.
Advanced → FlexConnect → FlexConnect Local Switching: Enabled, FlexConnect Local Auth: Enabled

Apply → Save Configuration

9. Configure the Wireless Host to connect to the WLAN
 - a. Create a new wireless profile on the host. Use the name **SMNET** for the profile.
 - b. Configure the profile for the SSID of the **SMNET** WLAN.
 - c. Use enterprise authentication with a username of **umme** and password of **salsabil**.
 - d. When you are finished, click Connect to Network. It will take time for the connection to be established.
10. Connect the Smart Phone to the **SMNET-GUEST** WLAN.

Part 8: Verification

At the end of the configuration, you should be able to browse www.google.com from all branch computers including wireless hosts.